

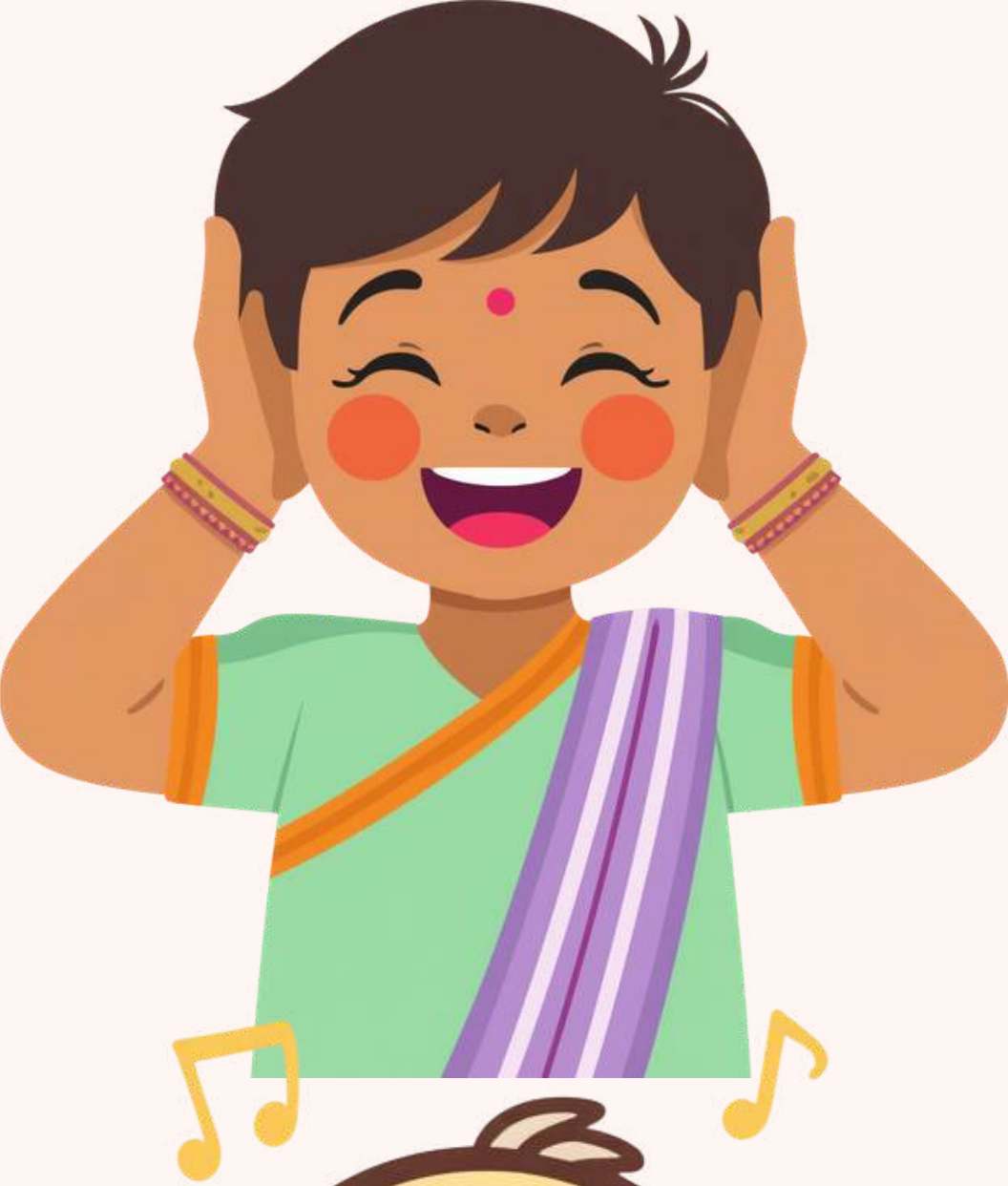


Why Does Sound Make Things Shake?

The Vibration Experiment
Planrs Academy



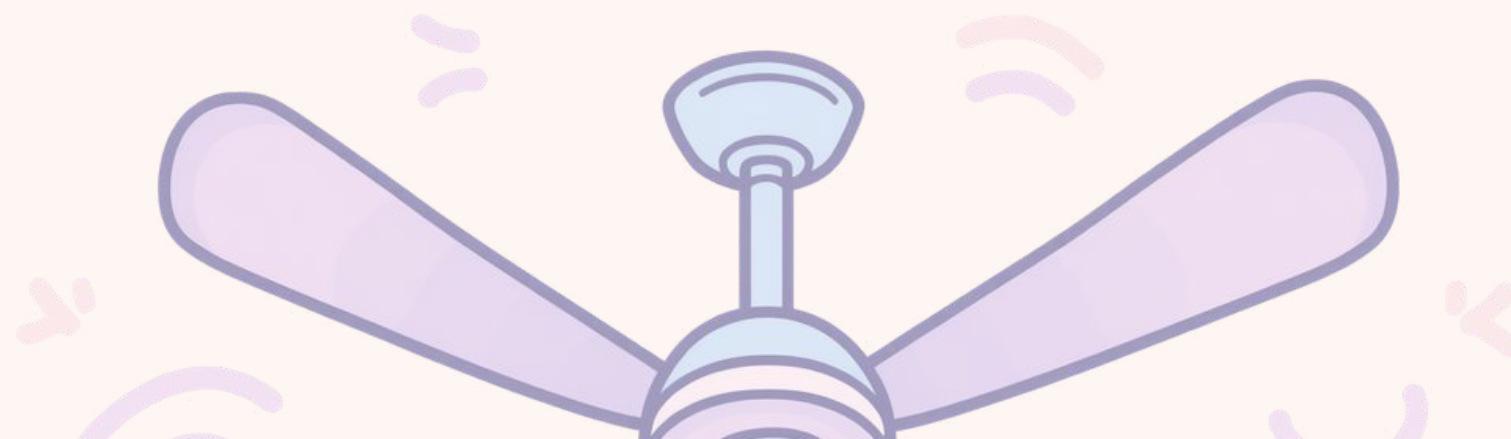
Sound Hunt Warm-Up



- Close your eyes and listen!
- What sounds can you hear around you?
- Birds chirping, ceiling fan whirring, traffic noise, school bell ringing, children laughing and talking



- Think about all the sounds you heard
- Where do sounds come from?
- Every sound has a source - something that makes it!



Meet Riya's Story

- It is Pongal festival time! Riya is at her grandparents' house in Tamil Nadu. She loves watching the colorful kolam designs and hearing the festive music everywhere.
- Riya picks up her grandfather's dholak drum and gives it a gentle tap. The drum makes a loud "dhum dhum" sound!
- Riya notices something strange - the drum skin is shaking! She looks up at her grandfather with big curious eyes.
- "Thatha, why does the drum shake when I hit it?" she asks. "And why does it make sound?"
- Thatha smiles. "That is a wonderful question, Riya! Let us find out together!"



What Is Vibration?

Vibration means shaking back and forth very fast!
Imagine a guitar string wiggling super quickly, or a dholak drum skin
bouncing when you hit it!

You can see vibration everywhere around you:

- Guitar string moving
- Dholak skin shaking
- Motorbike engine humming



Vibration Makes Sound

- When objects vibrate, they make sound!
- Think about it: when you pluck a rubber band stretched over a steel dabbaa, it shakes and makes a twanging sound.
- When someone plays the tabla, the skin vibrates up and down very fast – that's what creates the beat!

- A sitar string moves back and forth when plucked – that movement is vibration, and it makes beautiful music!
- Remember this simple rule:
Shaking = Vibration = Sound





Experiment 1 – Rubber Band and Dabbaa



Materials You Need:

- One rubber band (medium size)
- One steel dabbaa (tiffin box)

Steps to Follow:

- Stretch the rubber band over the open dabbaa
- Pluck the rubber band gently with your finger
- Listen carefully to the sound it makes!



Observation Time:

- What do you see? Watch the rubber band shake!
- What do you hear? A buzzing or humming sound!
- The rubber band vibrates and makes sound waves

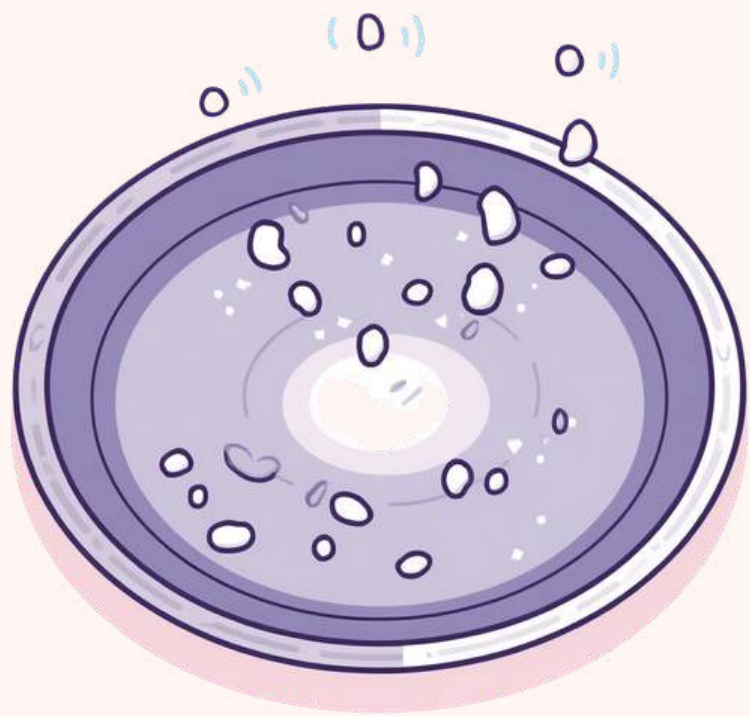
Remember: Vibration = Sound!



Experiment 2 – Salt and Thali

- Materials: Steel thali (plate), salt or sand grains
- Steps: Sprinkle a little salt on the thali
- Tap the plate gently OR speak loudly near it
- Watch what happens to the salt grains!

- Observation: The salt grains jump and dance!
- Why? Your voice or tap makes the thali vibrate
- The vibration makes the salt move around
- "Watch the salt grains move!"





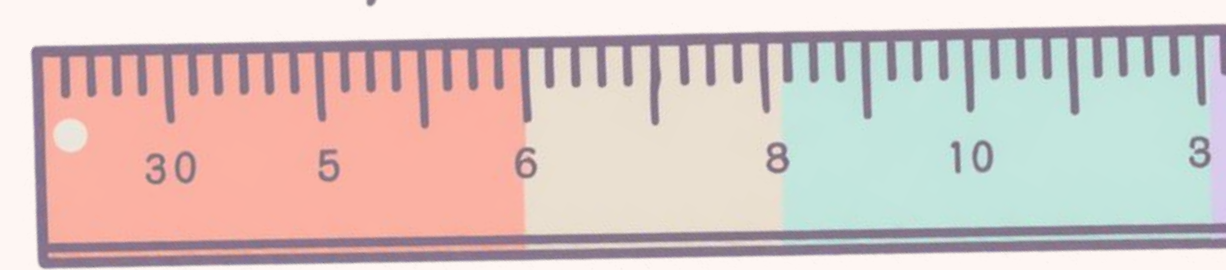
Experiment 3 – Ruler Vibration

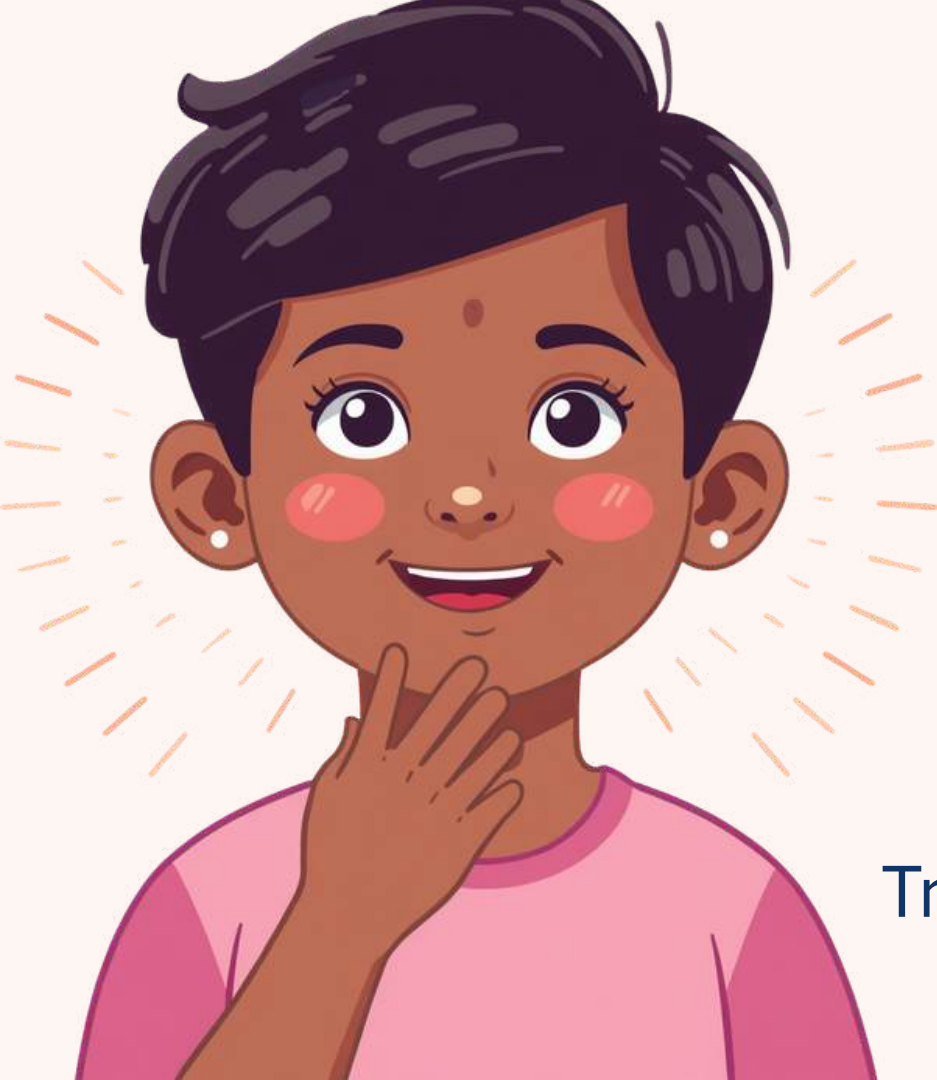
Materials You Need:

- A 30 cm ruler (scale)
- Your desk or table edge
- Steps to Follow:
 1. Hold the ruler firmly on the desk edge
 2. Let half of it hang over the edge
 3. Push the free end down and let go!

What to Observe:

- Watch the ruler shake back and forth
- Listen carefully – do you hear a buzzing sound?
- The faster it shakes, the higher the sound!
- Remember: The shaking ruler is vibrating and making sound!





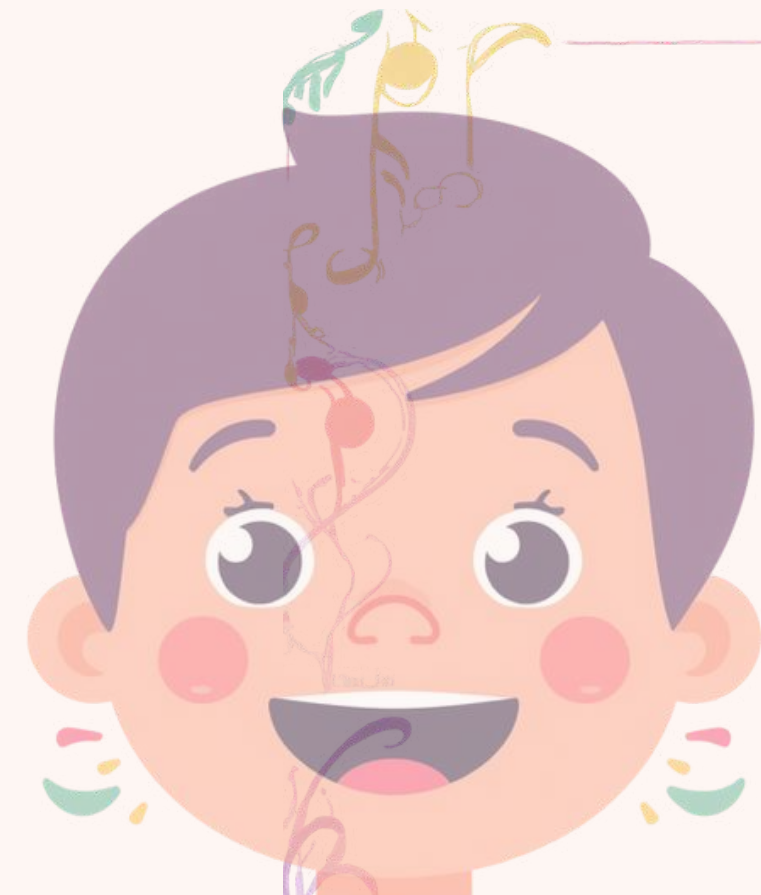
Feel Your Own Vibration



Try this fun activity! Place your fingers gently on your throat. Now say "Aaaa..." or "Ommm..." in a loud voice. Can you feel something shaking inside your throat? That shaking feeling is called vibration!

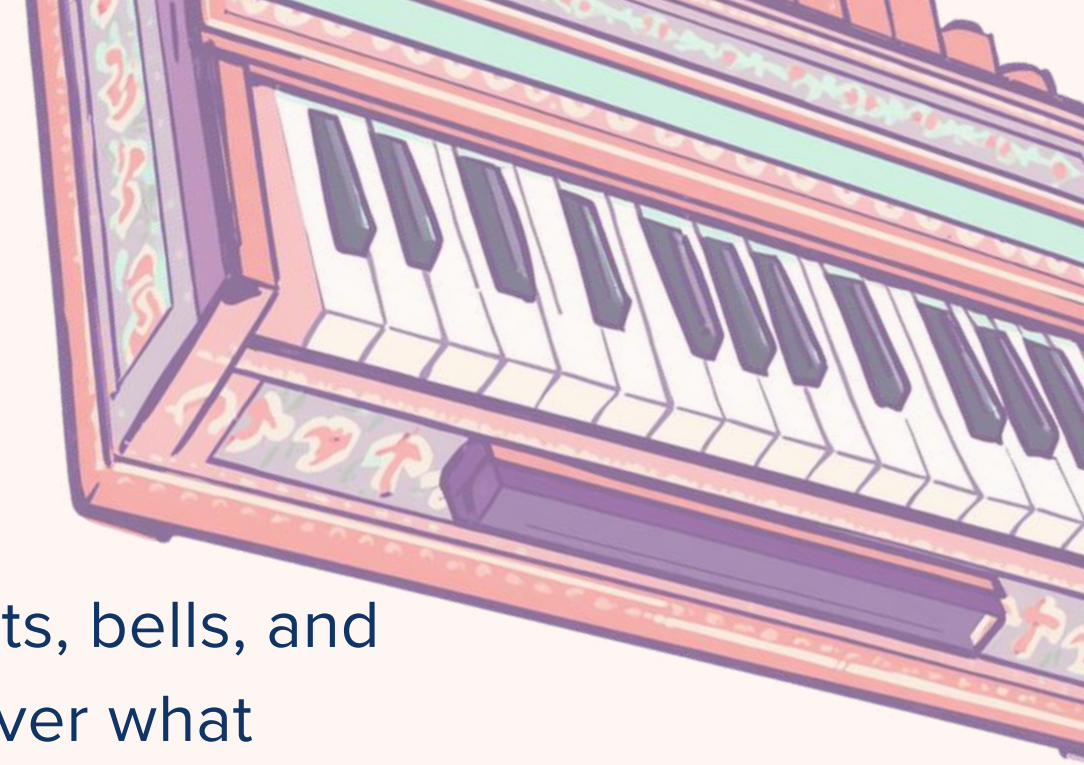


When you speak or sing, your voice box vibrates very fast. This vibration makes the sound of your voice! Try saying different sounds like "Mmmmm" or "Eeee" and feel how the vibration changes. Your body makes sound just like a musical instrument!





Sounds Around Us



So many things around us vibrate to make sound! Musical instruments, bells, and everyday objects all shake to create the sounds we hear. Let's discover what vibrates in each one!

- Tabla/Dholak – skin vibrates when you tap it
- Sitar/Veena – strings vibrate when plucked
- Bansuri – air vibrates inside the flute
- Harmonium – reeds vibrate when you press keys
- Temple bell – metal vibrates when struck
- School bell – rings and shakes
- Auto-rickshaw horn – beeps with vibration
- Pressure cooker whistle – steam makes it vibrate!





When Vibration Stops, Sound Stops

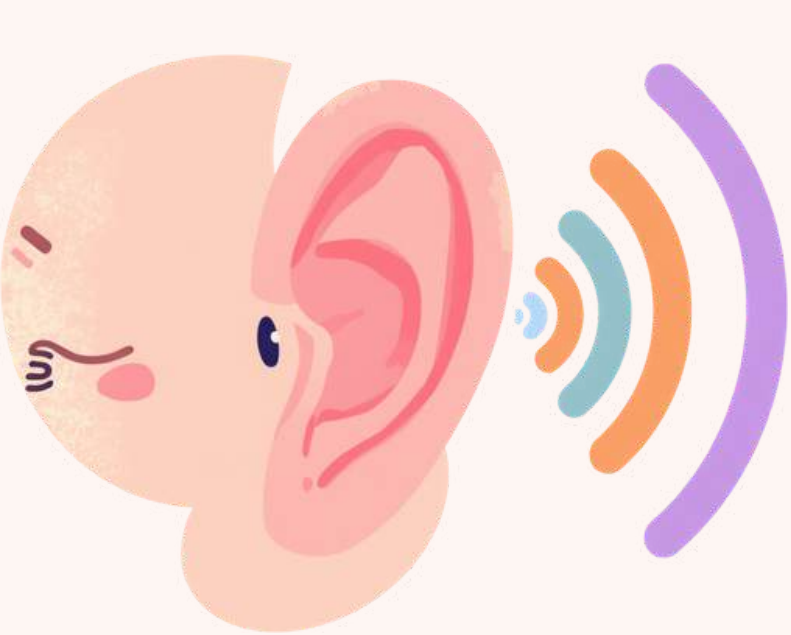


- No Vibration = No Sound!
- When something stops shaking, it stops making sound too.
- Try this: Hit a steel thali and watch it vibrate. The thali shakes and makes a ringing sound!



- Now Touch the Thali!
- Gently touch the vibrating thali with your hand. What happens?
- The shaking stops... and the sound stops too!
- "Touch the thali and see the sound stop!"



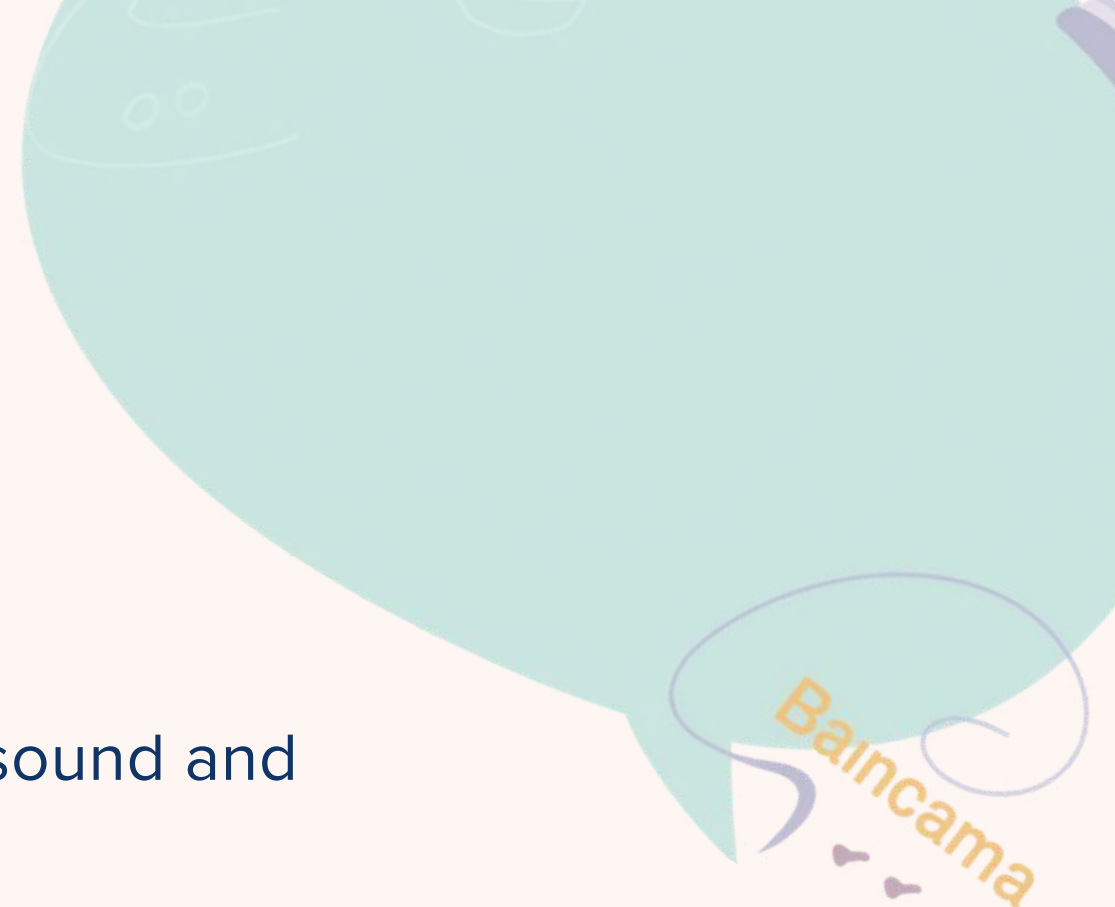
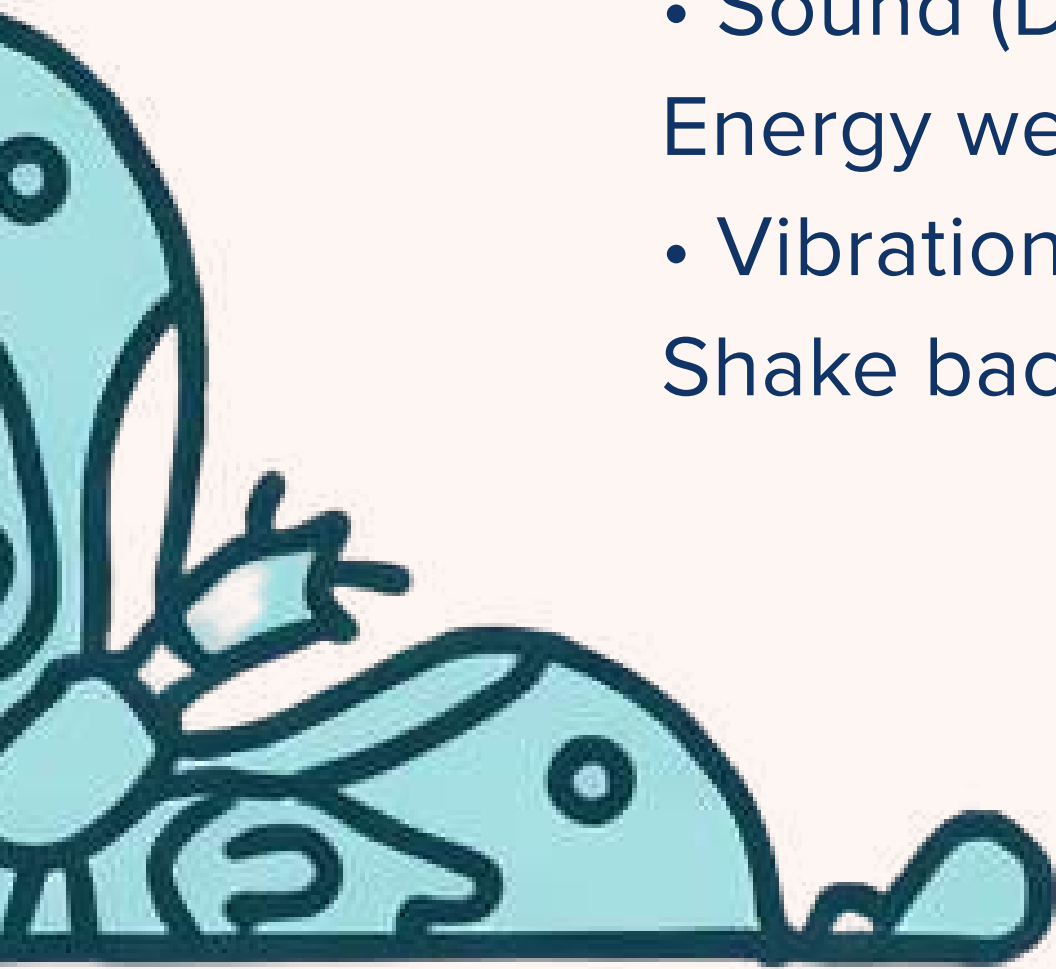


Vocabulary Review

Let's learn these important words! These words help us talk about sound and vibration. Say them out loud and remember what they mean!

- Sound (Dhwani)
Energy we can hear
- Vibration (Kampan)
Shake back and forth very fast

- Object (Vastu)
Thing that makes sound
- Produce / Make (Banana)
Create sound





Quick Quiz

Q1: What is shaking back and forth called?

✓ Answer: Vibration

Q2: What happens when vibration stops?

✓ Answer: Sound stops

Q3: True or False – A drum makes sound without being hit.

✗ Answer: False

Q4: Can you feel vibrations?

✓ Answer: Yes, in your throat when humming!





My Sound Journal

Draw one thing that makes sound! Think about objects around you - a rubber band being plucked, a tabla being played, or a bell ringing. Use the space below to draw your favorite sound-making object.



Now complete this sentence about your drawing:
"The _____ makes sound because it _____."
Example: "The tabla makes sound because it vibrates when I hit it!"

A rectangular box with a decorative border featuring stars, swirls, and musical notes. Inside the box are four horizontal lines for writing.



Homework – Sound Hunt at Home!



- Find ONE object at home that makes sound
- Touch it while it makes sound – feel the vibration!
- Examples: pressure cooker whistle, mixer grinder, doorbell, TV speaker



- Draw or write about your sound discovery
- Remember: If it vibrates, it makes sound!
- Share your sound hunt with the class tomorrow!





Thank You!
See You Next
Time!

